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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 13: EIA Processes - II

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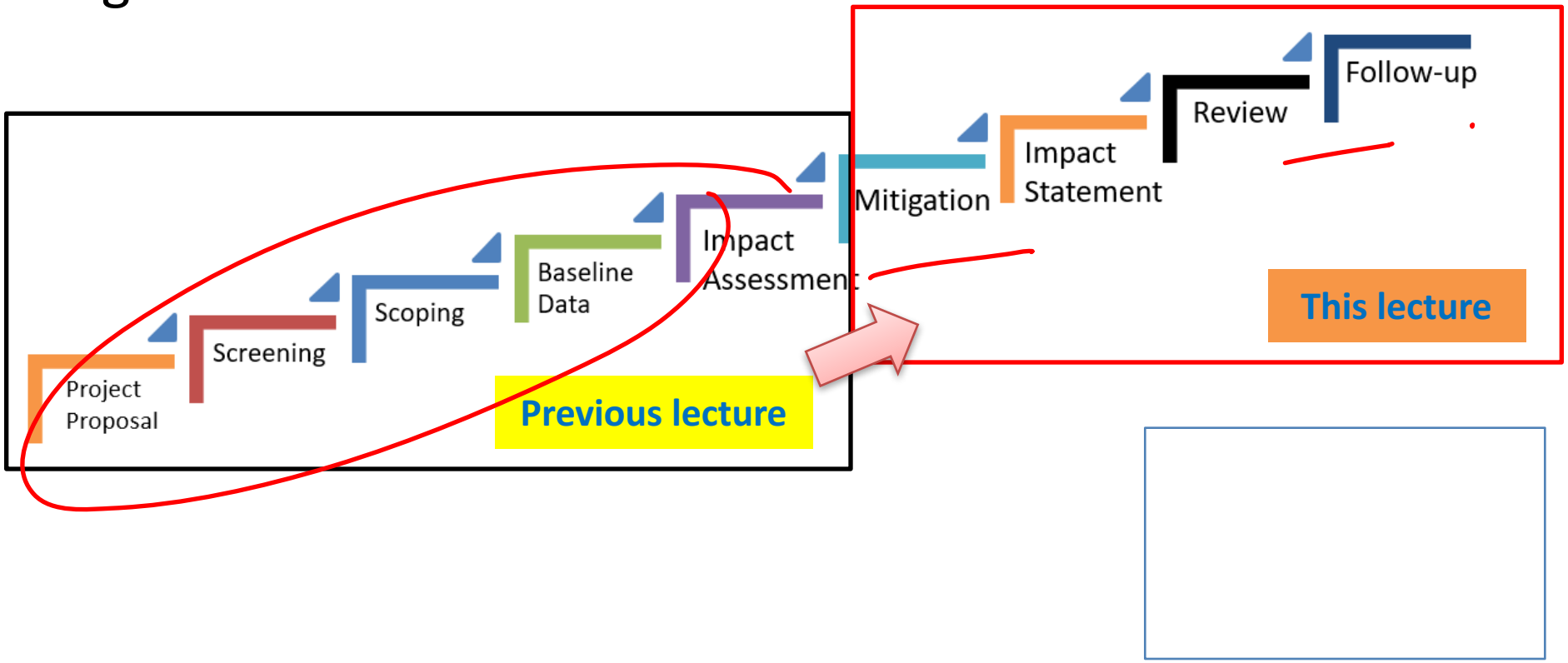


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- Steps of mitigation process
- Impact Management
- Contingency Plans
- Terms of reference (ToR)
- Environment Impact Statement (EIS)
- Review & Consultation including public participation



Stages of EIA



Mitigation process in EIA

- The Impact assessment stage in EIA reveals the possible damaging effects upon the environment.

- Mitigation process in EIA involves taking necessary measures to reduce or remove adverse environmental impacts.



Ex.: Construction of highways with Roadside plantations to reduce noise pollution in neighboring areas, limit the dust and increase the green cover which further helps in maintaining the better air quality, e.g., National Green Highway Mission.

Image Source: projectreporter.com

What are Mitigation Measures?

- Preventive measures that avoid the occurrence of impacts and thus avoid harm or even produce positive outcomes.
- Measures that focus on limiting the severity and the duration of the impacts.
- Compensation mechanisms for those impacts that are unavoidable and cannot be reduced further.

Ex.: Projects in wildlife sensitive area may impact local ecology but preserving local flora and fauna may result into better growth at another similar/controlled environment like Project Tiger (Govt. of India).



Mitigation Process: Objectives

To Identify the mitigation actions that mostly reduce the impacts, and that are practical and feasible to integrate into the planned alternatives.

To identify mitigation actions that reduce the severity of the impacts to the lowest possible level over the lifetime of the project and to avoid irreversible and cumulative effects.

To include activities designed to monitor compliance with agreements established in the assessment

To identify potential risks of accidents, malfunctions and other emergencies that may occur during all the project phases

Example of Project Tiger by the GOI



- Rapid decrease in population of tigers observed due to rapid urbanization and development activities around natural habitats.
- Project Tiger was launched in 1973 for tiger conservation.
- Positive results were noticed in the later years.

Image: First time population of tigers has seen growth after collaborative efforts by govt. & civil society

Source: Corbet-nationalpark.com

Example of Project Tiger by the GOI

RISE SPANS MOST STATES WHERE THE BIG CATS ARE FOUND

Change in tiger counts in some States and reserves

	2010	2014
Sunderbans	70	76 ↑
Northern Bengal	—	3
Uttarakhand	227	340 ↑
Uttar Pradesh	118	117 ↓
Bihar	8	28 ↑
Andhra Prasesh	72	68 ↓
Chhatisgarh	26	46 ↑
Madhya Pradesh	257	308 ↑
Maharashtra	169	190 ↑
Odisha	32	28 ↓



	2010	2014
Rajasthan	36	45 ↑
Jharkhand	10	3 ↓
Karnataka	300	406 ↑
Kerala	71	136 ↑
Tamilnadu	163	229 ↑
Goa	—	5 ↑
Assam	143	167 ↑
Arunachal Pradesh	—	28 ↑
Mizoram	5	3 ↓

Source: Corbet-nationalpark.com

Impact of Mitigation process on the Environment



Source: BBOP, 2009



Example of National Green Highway Mission



- Promote greening and development eco-friendly National Highway corridors across the country with participation of farmers, private sector and government institutions including Forest department.
- Address the issues that lie in the road of development and pave the way towards sustainable development

Source: morth.in
Image Source: medium.com

Example of Green Cover/Plantation

- Replanting of vegetation/green cover on slopes after road construction.
- Using trees to limit noise, dust around housing project.





Image Source: terramgeosynthetic.com

Example of Noise barriers along Railway tracks



Image Source: gaurdrialbarrier.com

- Noise barriers along the rail lines.
- Applicable specially if rail passes through populated area.
- Delhi metro has used it.

Example of Wildlife passages



- Help animals or other wildlife in navigating.
- Maintains natural drainage
- Facilitates local population for accessing nearby locations

Image Source: https://www.theguardian.com/environment/2021/jan/23/how-wildlife-crossings-are-helping-reindeer-bears-and-even-crabs-aoe

Example of Bamboo/Rope Bridge for Reptile Animals



Image Source: indiatimes.com

- Innovative approach implemented in Indian state of Uttarakhand.
- A bridge is created for movement of reptile creatures such as snakes, lizards etc.

Example of Smog Towers in Polluting areas



A Smog Tower in Delhi

Image Source: newindianexpress.com

- In case of any project affecting local air quality severely, then smog tower could be used to reduce the pollutants in the ambient air.

Example of Flora Conservation

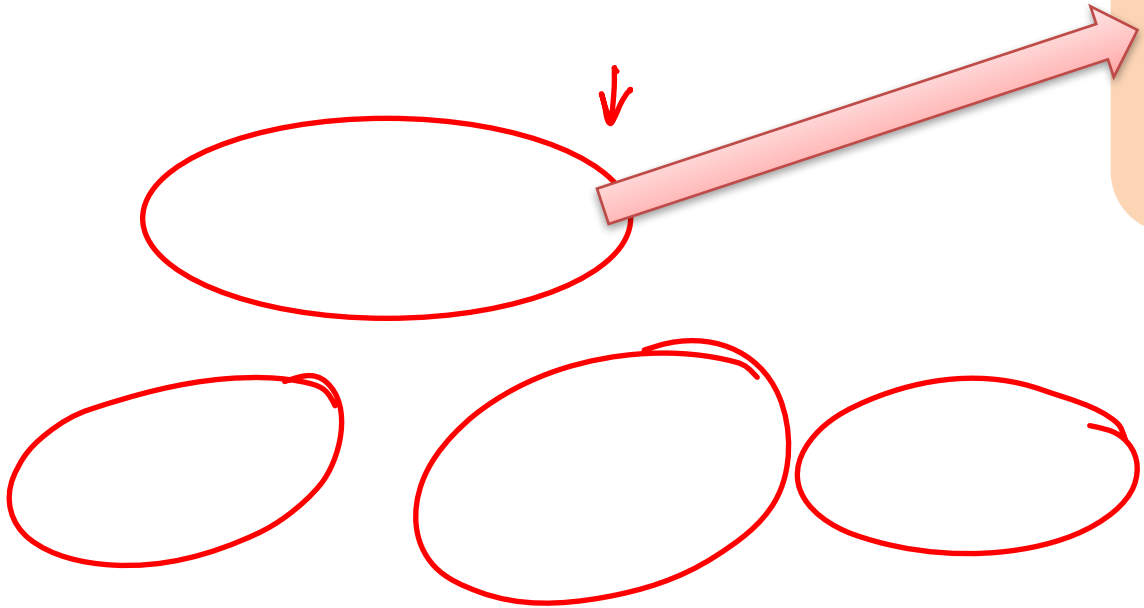


- Conservation projects are sub-part of EIA mitigation process.
- It helps in maintaining bio-diversity and conserving endangered plants & animals.

Image: The Western Ghats of India are the most diverse tropical region in the world

Image Source: m.economictimes.com

Impact Management Plan (IMP)



Impact management plan is the creation of a series of plans and protocols aiming to manage and monitor the identified mitigation measures and risks.

Objectives of Impact Management Plan (IMP)

- Plan to implementation of mitigative measures.
- Additional action plans to manage risks and carry out monitoring will also be required.

IMP depends on

- Type of project
- Identified impacts
- Risks associated
- Natural and social context



Elements of Impact Management

Mitigation measures

Summary of all mitigation measures and details on how these will be implemented; the measures are linked to the impact analyses

Monitoring

The plans for monitoring the environmental impacts and the effectiveness of mitigation measures in addressing the impacts

Capacity Development

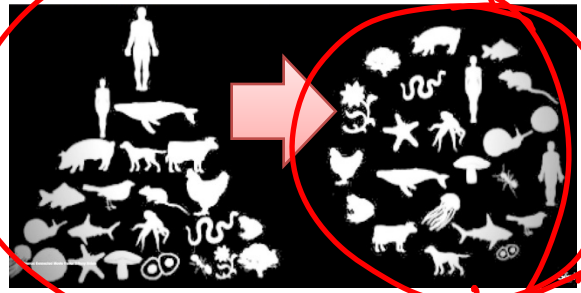
The assessment of capacity-building needs required to involve stakeholders in managing environmental and social impacts and monitoring efforts

Implementation Schedule and Cost Estimates

Costs of the outlined mitigation activities, capacity-building and monitoring; this will include cost of implementation and then ongoing monitoring and follow-up capacity development costs

Integration

The EMP needs to be developed in a way that fits with the planned project, the mitigation actions, and other activities linked to the project



Resettlement & Rehabilitation(R&R)

- To minimize or reduce the number of people affected and displaced by a project.

- Close collaboration with the local authorities such as municipal leaders and key agencies.

- Processes in place to reach out to the community and establish good connections with community leaders.

Rehabilitation & Resettlement (R&R)

- In case of change in livelihood, it prepares for new kind of jobs.
- Prepares for living in new region where resettlement is done.
- Overall provide similar or better lifestyle to affected population.



Image Source: ntpc.co.in

Contingency Plan

Plan must include an assessment of

- Possible risks and external contingencies (natural events and disasters)
- Identifying and determining those potential risks or threats to the health of the population and the structure of ecosystems

Definition:

A contingency plan clearly establishes lines of action to follow, assigning responsibilities, functions and establishing quick and effective responses to emergency situations



Example of Contingency Plan

- Protocols in case of industrial accidents or fires.
- Preparation and safety measures for emergencies like flood, earthquake etc.





Image Source: hsewatch.com, downtoearth.org.in

Example of Bhopal Gas Tragedy



Image: Industrial plant of Union Carbide corporation where gas leak occurred

Image Source: deccanherald.com

- In 1984, highly toxic Methyle Isocyanate (MIC) gas leaked from pesticide plant.
- 3787 people died and 574336 severely affected.
- A proper contingency plan and better safety measures could save many lives.

Environmental Impact Statement (EIS)

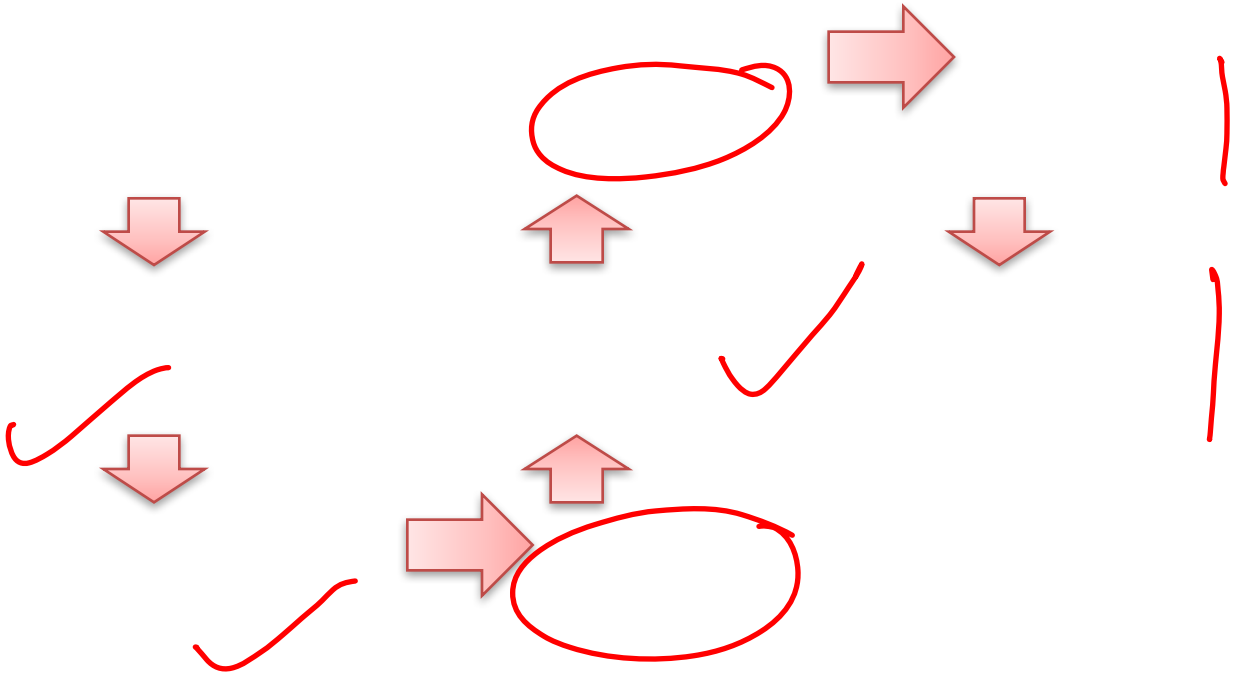
Environmental statement (EIS) is defined by the International Chamber of Commerce as a management tool comprising a systematic, documented, periodic and objective evaluation of how well environmental organizations, management and equipment are performing with the aim of helping to safeguard the environment. This is the final stage of an EIA.

An Ideal content of an EIS should have the following:

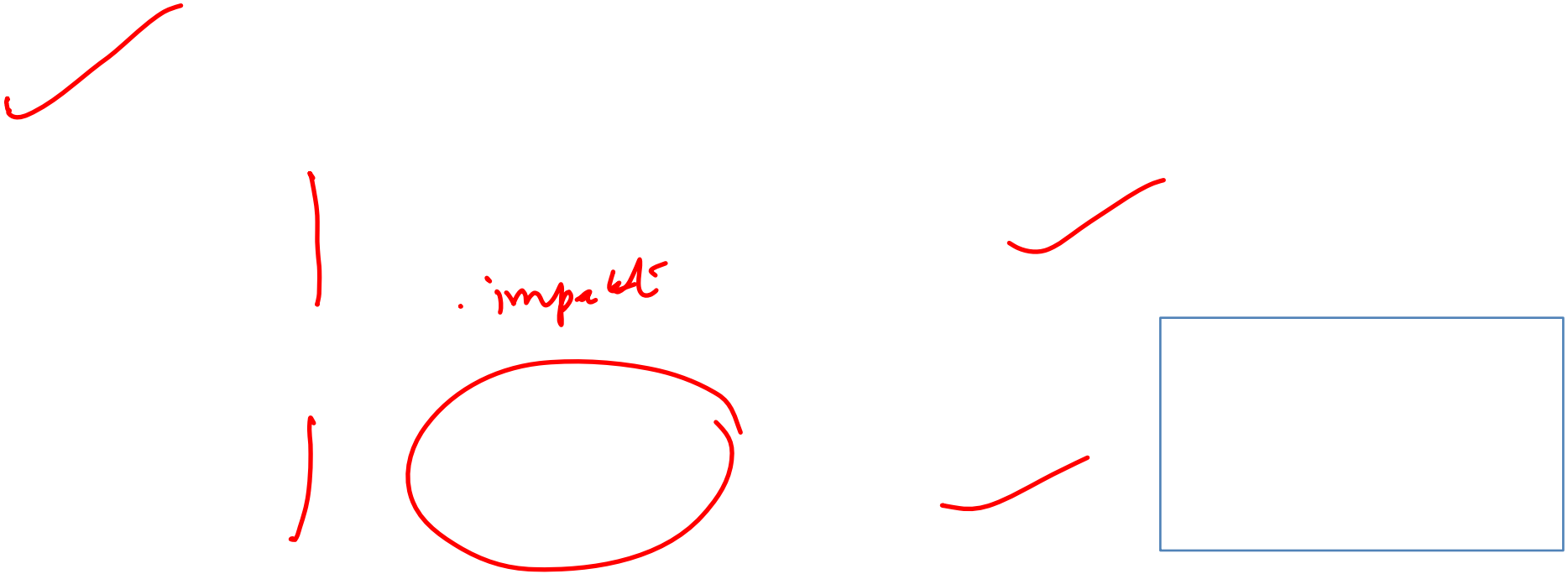
- Executive summary
- Policy, legal and administrative framework
- Description of the environment
- Description of the proposed project in detail
- Significant environmental impacts
- Identification and analysis of alternatives
- Mitigation action/mitigation management plan
- Environmental management plan
- Monitoring program
- Knowledge gaps
- Public involvement
- List of references
- Appendices including reference documents, photographs and unpublished data
- Terms of Reference
- Consulting team composition
- Notes of public consultation sessions.



Format of EIS Report: Inter-American Development Bank (IDB)



Format of EIS Report: European Union (EU)



Format of EIS Report: World Bank



Environmental Impact Statement (EIS) Review

EIS review is done by competent authority as per legal requirements.

In India: It is done by the government (MoEF).



Objectives of EIS Review

1.

2.

3.

4.

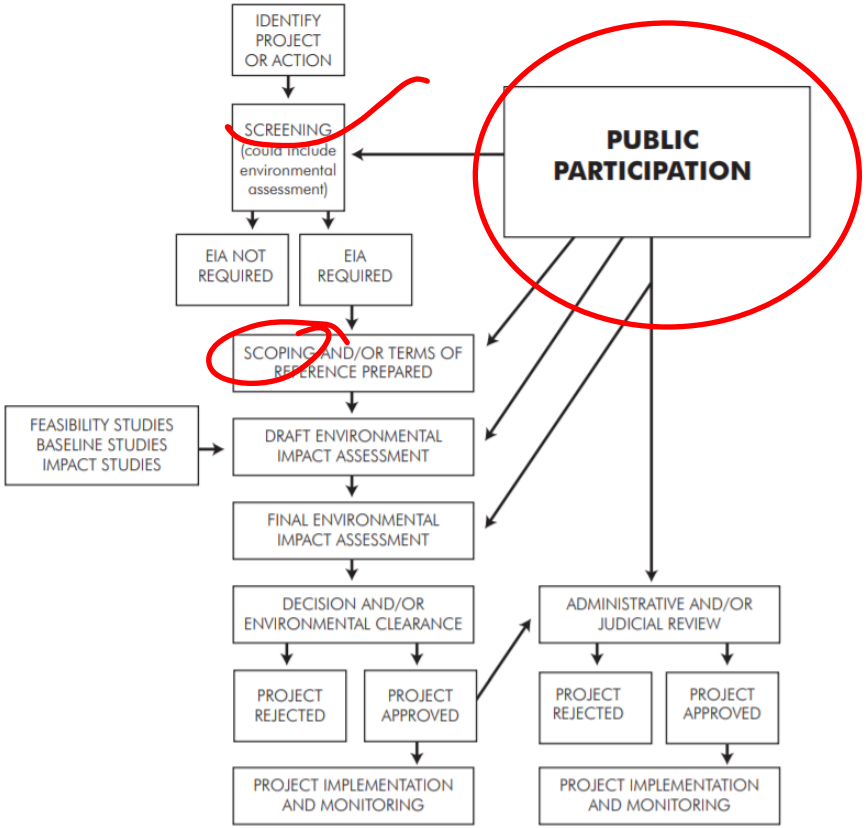


Public Participation / Consultation

The overall goal of public Participation is to engage key stakeholders' groups such as citizens, NGOs, agencies, authorities and interest groups to provide their input into the planned development and especially on those impacts that directly or indirectly affect people's livelihoods



Public Participation at various Stages of EIA



In practice, public consultation is done mainly during review stage.

Source: Guidebook for evaluating mining project EIAs

Features of Effective Public Participation

Inform and involve interested and affected stakeholders like citizens, public authorities and interest groups

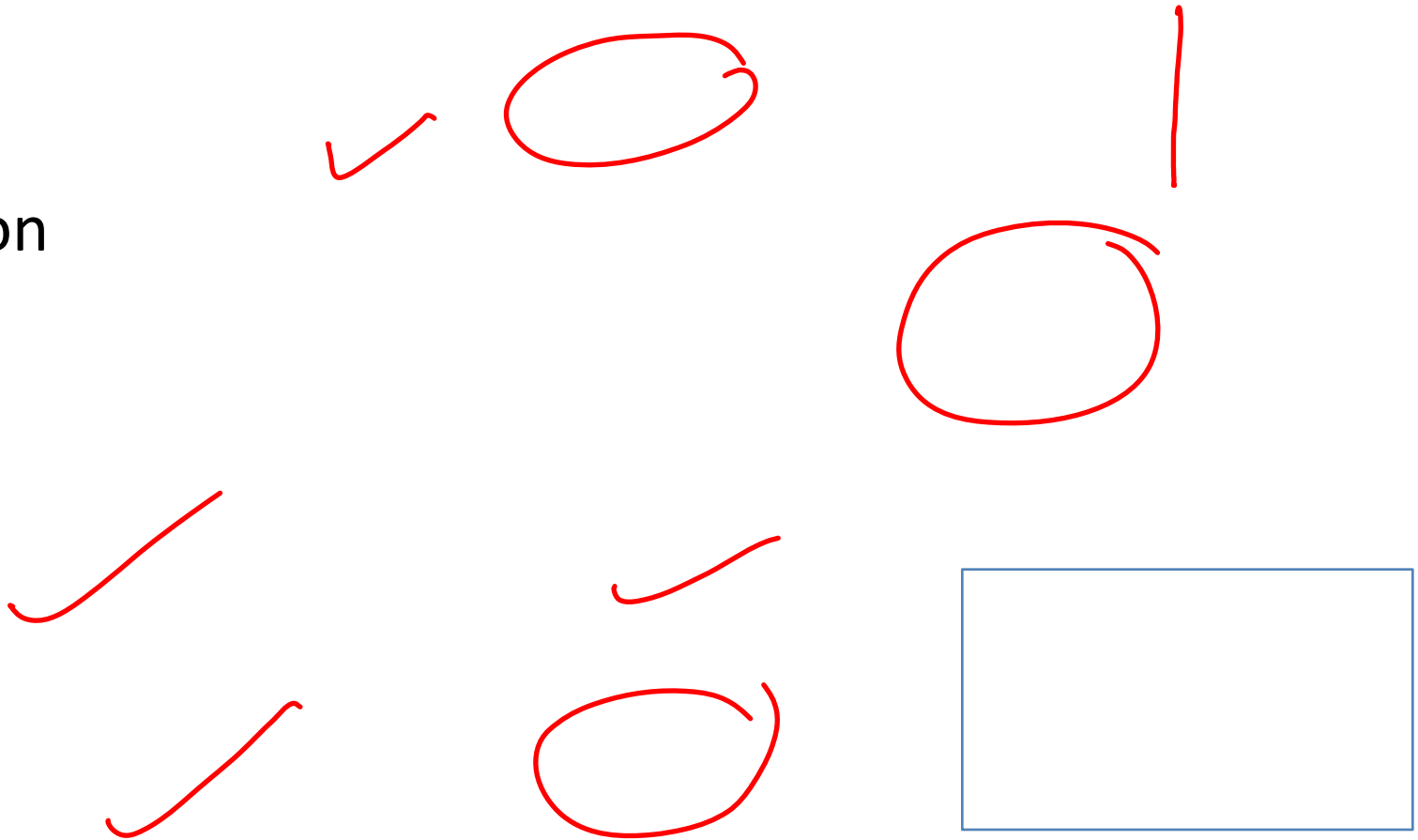
Explicitly address comments, concerns and inputs, both in documentation and decision making

Ensure that all relevant actors and parties are involved or at least represented

Safeguard sufficient access to all relevant documents and information concerning the project and the EIA documentation

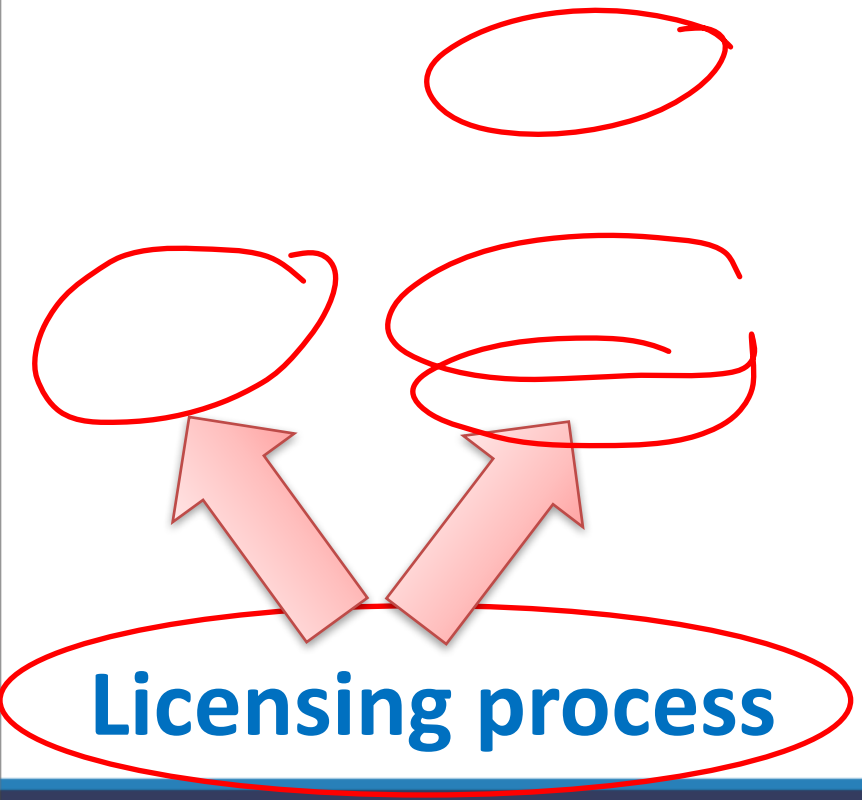
Source: Aschemann, 2004

Steps for Public Participation



Source: World Bank, 2010

Licensing



Licensing is the outcome of the review process.

Follow-Up/Monitoring

The Follow-up or Monitoring process ensures:

- Accuracy of the original predictions
- Degree of the deviation from the predictions
- Possible reasons for any deviation
- Whether mitigation measures have achieved their objectives in reducing or eliminating impacts

Information generated by this process can contribute to the improvement of future EIA practice, for example, by enabling more accurate predictions to be made.



Monitoring process: World Bank



Monitoring process: Asian Development Bank

Asian Development Bank (ADB) is an international organization which funds development projects across Asian countries.



Monitoring process: European Bank for Reconstruction & Development (EBRD)

EBRD (European Bank for Reconstruction & Development) is an international organization which funds development projects.



General Shortcomings of EIA Reports

Example EIA Report	Shortcoming
An EIA report describes the proposed construction of an industrial plant but omits information about construction of a pipeline and other facilities to transport and handle raw materials and finished products to and from the plant.	The description of the proposal does not cover key features.
An EIA report describes the proposed construction of a coal-fired power plant using surface water as a cooling medium. It does not divulge that the surface water body is already used by other industrial activities for this purpose to the limit of its cooling capacity.	Key problems affected by the proposal are not described.
An EIA report for a pipeline project does not indicate that the proposed alignment will bisect certain areas of ecological value.	Sensitive elements in the affected environment are overlooked.
An EIA report for a sanitary landfill indicates that the soil types in the area are very diverse, ranging from sand and clay to peat. The alternatives do not take into account the large differences in compaction and subsidence of these soil types, with subsequent failure of underlining and drainage systems.	Alternatives do not comply with environmental regulations and standards.
An EIA report for a sanitary landfill does not describe a system for collecting methane gas produced in the landfill, even though greenhouse gas emissions contribute to climate warming and should be capped at current levels.	Appropriate mitigating measures are not considered.
An EIA report for a sanitary landfill in an area with very variable soil conditions does not describe the environmental risks and consequences of a possible failure of the underlying sealing and drainage systems.	Serious environmental impacts or risks are not described or are incorrectly described.
An EIA report on an urban development scheme makes use of a mobility prediction model using national averages, although local data is available and would permit a more precise prediction to be made.	Insufficient or outdated prediction models are used.

Source: *Netherlands EIA Commission in UNU, UNEP, RMIT (2007).*

Conclusions

- An EIA process balances developmental & environmental aspects.

- In absence of EIA process, many harms could be done to environment and indigenous population. An EIA ensures right protections to society and environment.

- A proper implementation of EIA process, identifying impacts and following suitable mitigation measures ensures Sustainable growth and benefits to community.

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- Environmental impact assessment general procedures,

Thank You